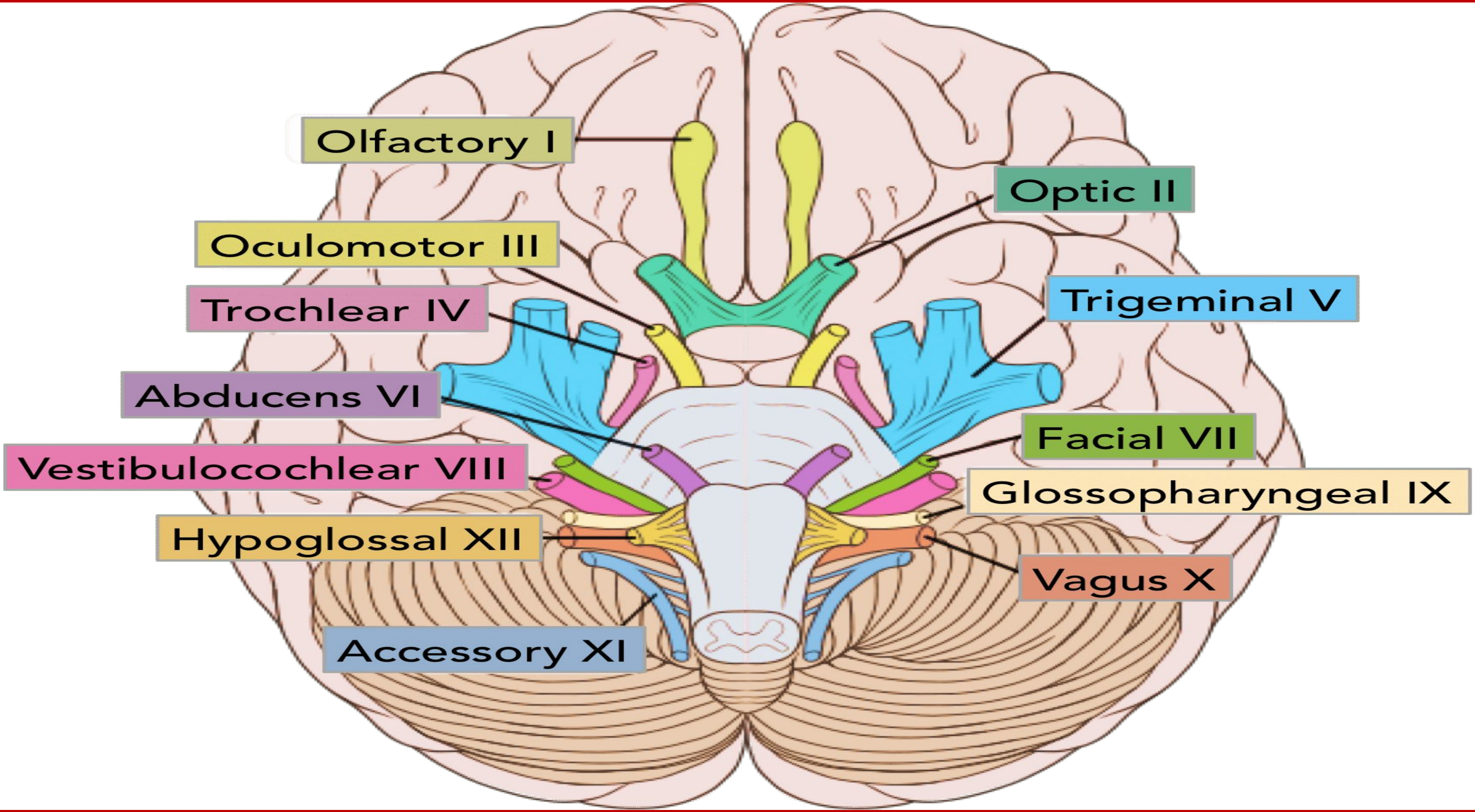
EXAMINATION OF THE CRANIAL NERVES



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Learning Objectives

By the end of the Practical, the student should be able to :

- Know the afferent, efferent and mixed Nerves.
- Explain the functions related to each cranial nerve.
- Demonstrate the functions of cranial nerves.
- Identify the defects, if any.

Theoretical Explanation

The Cranial Nerves are classified as follows:

- Purely Afferent (Sensory) I, II , VIII.
- Purely Efferent (Motor) III, IV, VI, XI, XII.
- Mixed V, VII , IX , X.

The Olfactory nerve (CN I)

- Simply tested by offering something familiar for the patient to smell and identify – for example **coffee or vinegar.**

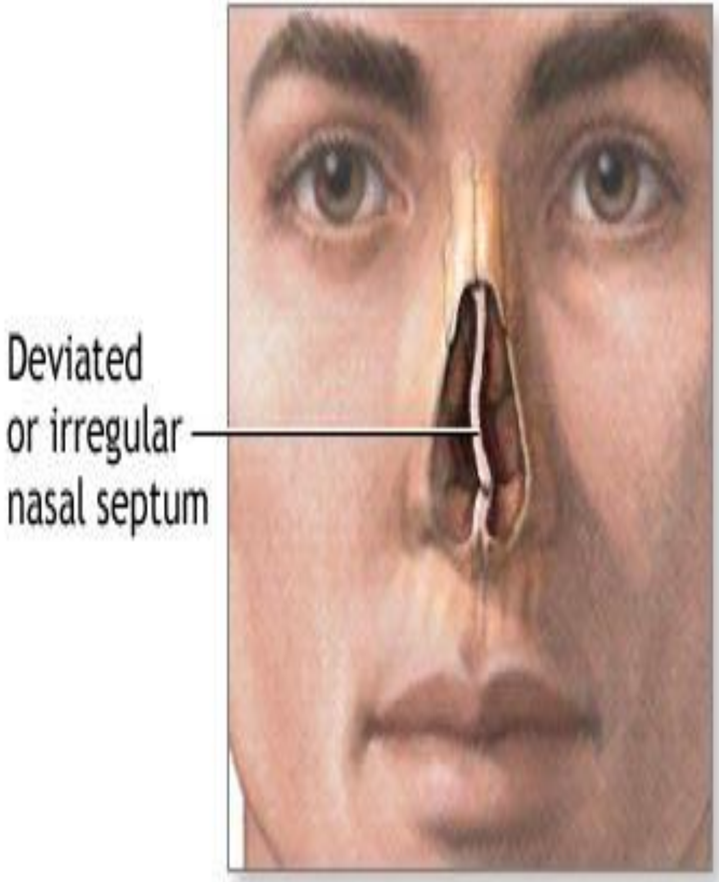
Abnormalities of Olfaction

- **Ansomia:** Loss of sense of smell.
- **Hyposmia:** Decreases sense of smell.
- **Hypersomia :** Increase sense of smell.
- **Perosmia:** Altered sense of smell.

Causes: Abnormalities of Olfaction

Unilateral loss imply a **structural brain lesion** affecting the olfactory bulb or tract.

- **local causes** such as deviated Nasal Septum or blocked nasal passage.
- **Bilateral loss** can occur with rhinitis or damage to cribriform plate.



Precautions

- Before experiment ask the subject to blow the nose properly.
- Check each nostril separately.
- Close the eyes of the subject.
- Use each substance separately to avoid mixing of smell of different substances.

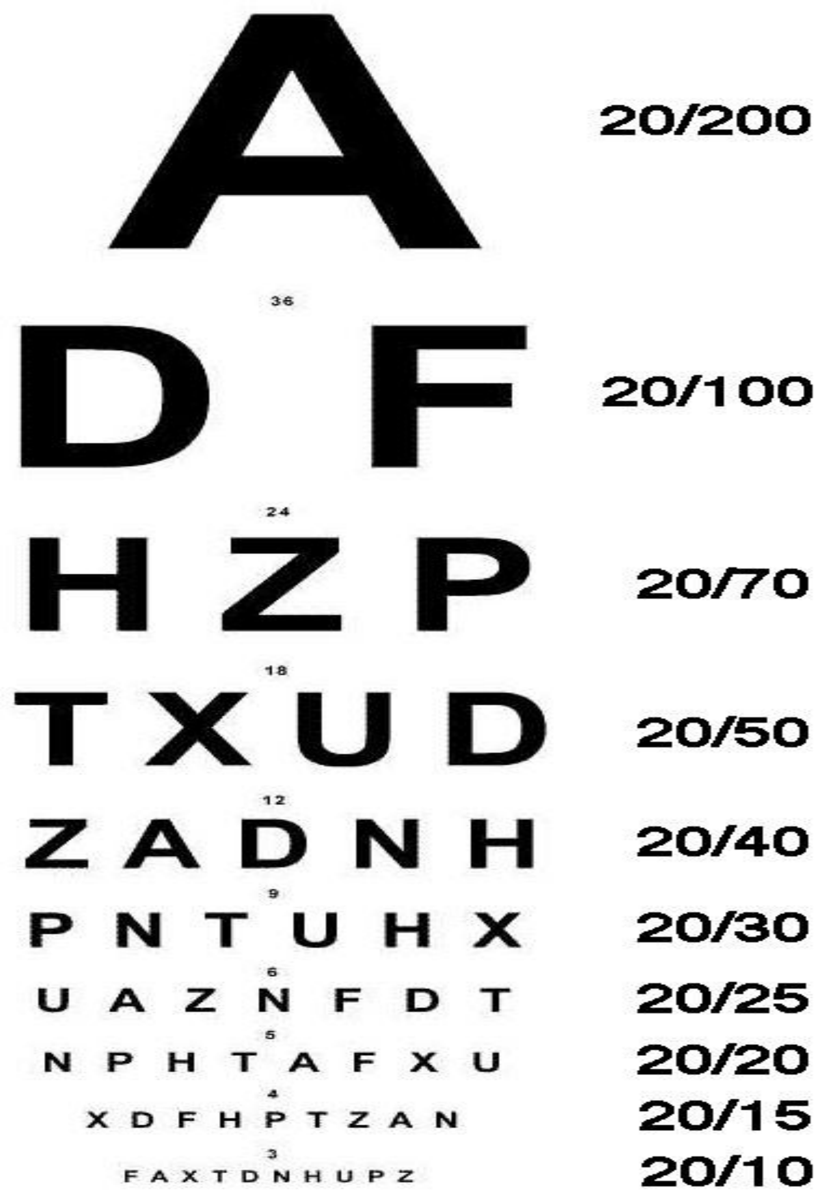
The Optic nerve (CN II)

Tested in five ways:

- Acuity
- Color
- Fields
- Reflexes
- Fundoscopy

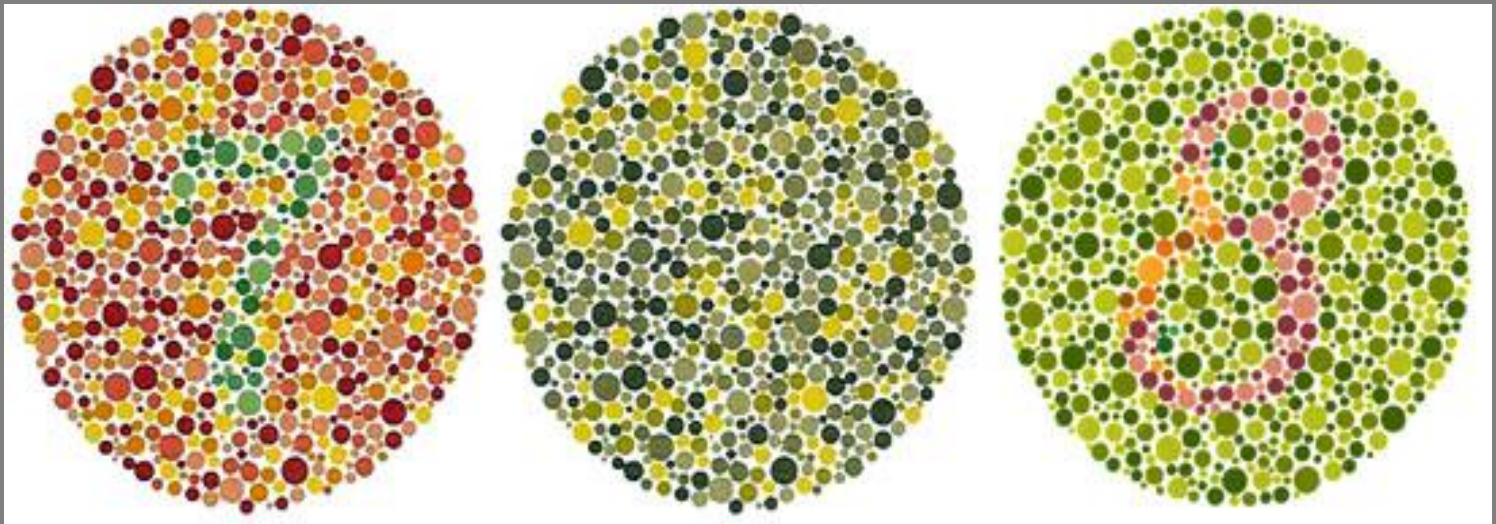
The Optic nerve (CN II) - Acuity

- The acuity is easily tested with **Snellen charts**. This should be assessed both with the patient wearing any glasses or contact lenses they usually wear and without them.
- The normal height for the letter A is 88 mm, and the viewing distance is 6 meters



The Optic nerve (CN II) - Color

- Colour vision is tested using Ishara plates, these identify patients who are colour blind.



The Optic nerve (CN II) - Fields

- Subject is allowed to sit comfortable in front of the examiner at a distance of 2 feet with one eye closed.
- Subject is asked to look at one object right in front of the eye and the eye should be fixed.
- Finger is moved in all directions(fields) i.e.
  - Nasal field,
  - Temporal field,
  - Superior field and
  - Inferior field ,

And the subject is asked if he can see.

- Procedure is repeated for the second eye.

The Optic nerve (CN II) - Reflexes

- Place one hand vertically along the nose to block any light from entering the eye not being tested.
- Shine a pen torch into one eye and check that the pupils on both sides constrict.
- This should be tested on both sides.

The Optic nerve (CN II) - Fundoscopy

- Finally Fundoscopy should be performed on both eyes.

The examination of the ocular fundus to assess retinal manifestations of hypertension, DM, retinal detachment, melanoma.

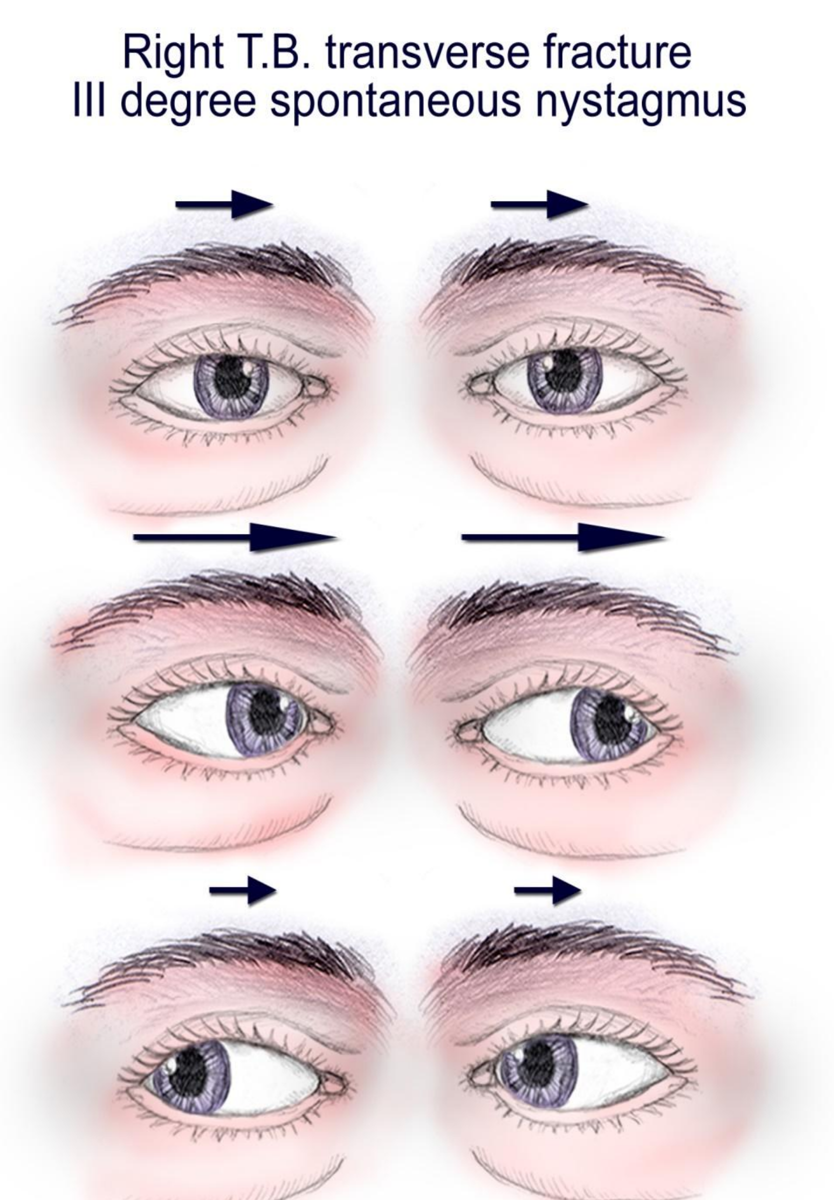
Occulomotor nerve (CN III), Trochlear nerve (CN IV), Abducent nerve (CN VI)

- These 3 nerves are examined together as they control the external ocular muscles involved in movements of the eye.

- Asking the patient to look to each side , up and down following an ‘H’ pattern keeping their head perfectly still directly in front of you.

Occulomotor nerve (CN III), Trochlear nerve (CN IV), Abducent nerve (CN VI)

- Ask the patient to follow a target such as your finger or a pen with their eyes without moving their head.
- Pause at the ends of each direction of gaze to observe for nystagmus.(Nystagmus is a vision condition in which the eyes make repetitive, uncontrolled movements.
- Assess saccadic eye movements by having the patient make quick horizontal and vertical eye movements.
- Eyeballs are observed for any oscillatory movements.



The Trigeminal nerve (CN V)

Examination:

Examine the motor and sensory systems.

- Involved in sensory supply to the face and motor supply to the muscles of mastication.

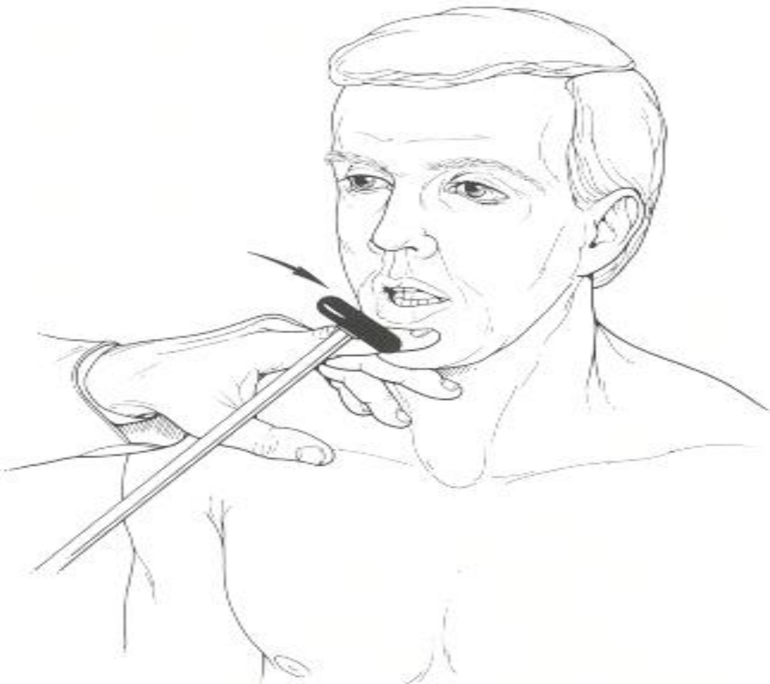
The Trigeminal nerve (CN V)

- Examining the sensory supply to the face

- Initially test sensory branches by lightly touching face with a piece of cotton wool and then with a blunt pin in three places on each side –
    - around the jawline,
    - on the cheek and
    - on the forehead.
- The corneal reflex should also be examined as the sensory supply to the cornea is from this nerve.
- This is done by lightly touching the cornea with the cotton wool.
- This should cause the patient to shut their eyelids.

The Trigeminal nerve (CN V)- Examining the motor supply

- Ask the patient to clench their teeth together, observing and feeling the bulk of the masseter and temporalis muscles.
- Then ask them to open their mouth against resistance.
- Finally perform the **jaw jerk** on the patient by placing your left index finger on their chin and striking it with a tendon hammer.
- This should cause slight protrusion of the jaw.



The Facial nerve (CN VII)

- Supplies motor branches to the muscles of facial expression.
- Therefore, this nerve is tested by asking the patient to crease up their forehead (raise their eyebrows), close their eyes and keep them closed against resistance, puff out their cheeks and show you their teeth.

The Vestibulocochlear nerve (CN VIII)

- Purely sensory nerve with two divisions:
  - Cochlear.
  - Vestibular.

Cochlear:

- For the Rinne test, place a sounding tuning fork on the patient's mastoid process and then next to their ear and ask which is louder, a normal patient will find the second position louder. For Weber's test, place the tuning fork base down in the center of the patient's forehead and ask if it is louder in either ear. Normally it should be heard equally in both ears.

The Vestibulocochlear nerve (CN VIII)

- Vestibular Nerve:
- Subject is asked whether he sees objects moving around hi m or he feels giddiness or dizziness.

The Glossopharyngeal nerve (CN IX)

- **Most of the 9th nerve is intermingled with the tenth nerve.**
- **Taste on the posterior 1/3rd of the tongue is difficult to test on the bedside.**

Sensory function (Sensory component : 9th Nerve).

Motor Function (Motor component is the 10th nerve).

The Glossopharyngeal nerve (CN IX)

Sensory function:

GAG REFLEX:

- Open the mouth and depress the tongue with a disposable spatula.
- Touch the posterior pharyngeal wall with a stick having cotton wrapped on end.
- First on one side of the midline and then the other.
- There will be contraction and elevation of the pharyngeal wall on that side.

Motor Function:

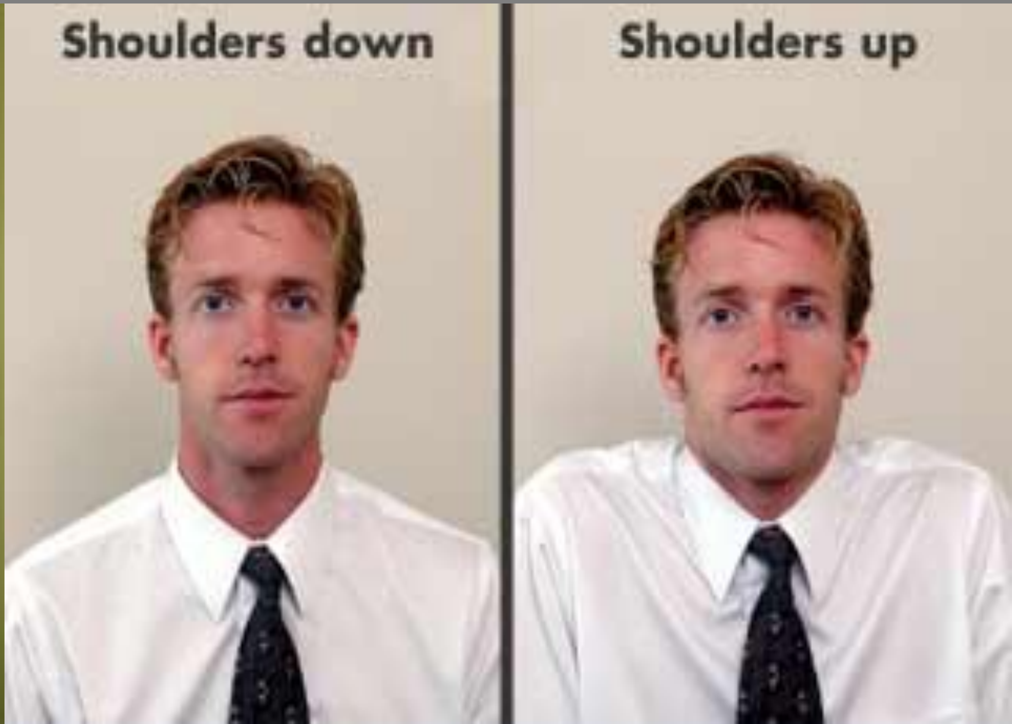
- Cannot be tested independent of the 10th nerve.

The Vagus nerve (CN X)

- Provides motor supply to the pharynx.
- **Speech:** if the larynx is paralyzed, there is dysphagia (hoarseness).
- If soft palate is paralyzed , voice has nasal quality.
- You should also observe the uvula before and during the patient saying '**aah**'. Check that it lies centrally and does not deviate on movement.

The Accessory nerve (CN XI)

- Gives motor supply to the sternocleidomastoid and trapezius muscles.
- To test it, ask the patient to shrug their shoulders and turn their head against resistance.



The Hypoglossal nerve (CN XII)

- Provides motor supply to the muscles of the tongue.
- Observe the tongue for any signs of wasting or fasciculations.
- Then ask the patient to stick their tongue out.
- If the tongue deviates to either side, it suggests a weakening of the muscles on that side.